# Bayesian Analysis

## Frequentist and Bayesian Approaches in Modeling the Energy Uncertainty Index

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# 1 Introduction

In recent years, energy has become one of the most discussed topics and the strategies aimed at securing energetic supply keep occupying the main pages of newspapers. The gas crisis following the Russian invasion of Ukraine and the conflict between United States and Iran to control to Strait of Hormuz are just some examples of the major disputes that have control over energy-related resources as the main goal.

This project investigates whether a Bayesian inferential framework delivers a superior forecasting performance than its frequentist counterparts, regarding energetic uncertainty. The question is not just technical: frequentist and Bayesian models differ not only in estimation philosophy but in how they communicate uncertainty.

To examine this, four models are estimated and compared. At the univariate level, a frequentist ARIMA(2,0,1) is compared against a Bayesian AR(3) with informed priors derived from OLS estimates, both applied exclusively to the log-transformed Energy Uncertainty Index ($log_eui$).
At the multivariate level, a frequentist VAR(2) is compared against a Bayesian VAR(2) with a diffuse prior, both incorporating geopolitical risk, climate policy uncertainty and oil prices as endogenous variables.

# 2 Data Exploration and Preprocessing

The original datasets span different time periods. To ensure consistency across all variables, the analysis was restricted to the common period 1996–2023, resulting in 335 monthly observations.
The target variable is the EUI (Energy Uncertainty Index), which considers 28 countries and for each one, the index is computed by counting the frequency of energy-related terms in national newspapers. These country-level uncertainty measures are then weighted by GDP to form an aggregate series, which is finally normalized to have a mean of 100 over the sample period.
Three influential factors are considered and used as explanatory variables: GPR, CPU and Europe Brent Spot Price. GPR (Geopolitical Risk Index) is calculated by counting the share of articles containing keywords related to adverse geopolitical events in 10 globally famous newspapers each month. The results are obtained by an automated text search on the electronic archives of these newspapers. CPU (Climate Policy Uncertainty) is built similarly to the GPR; therefore, the authors

consider 8 newspapers and the share of relevant articles per month and then they standardize it. The Europe Brent Spot Price is the price of crude oil extracted in the North Sea and it reflects worldwide supply–demand conditions and reacts sharply to geopolitical tensions and major energy shocks. For simplicity, this variable was labeled *oil_price*.
The variables exhibit non-stationary trends and differing scales, which was addressed by applying a log-transformation to stabilize variance.
Afterwards, the correlation pattern between variables must be checked (Figure 1) since it is crucial to address the causal relationship between the relevant variables. The variables present a good level of correlation, meaning that treating any single variable as strictly exogenous would be inappropriate, as the direction of causality is not clear a priori. To account for this endogeneity, a VAR framework is adopted.

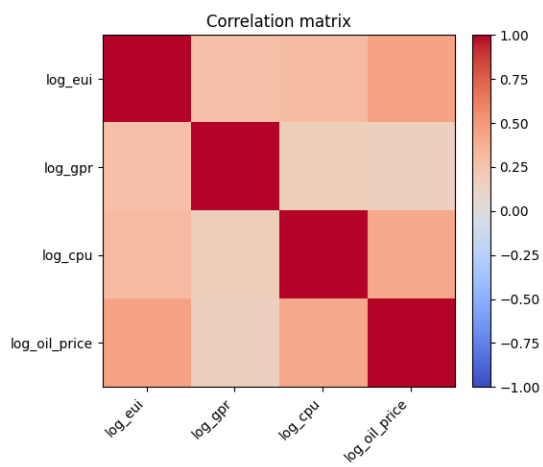


Figure 1: Correlation matrix.

The ACF/PACF plots indicate that non-stationarity is present but not severe across any of the variables. Note that all the variables are considered on the log-level for coherence with the target.

# 3 Models Implementation

## 3.1 Frequentist Models

An ARIMA(p,d,q) model is defined by:

$$\phi(L)(1-L)^d y_t = \theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2)$$

where:

1. d is the number of differences applied to make the series stationary
2. $\phi(L) = 1 - \phi_1 L - \ldots - \phi_p L^p$ is the AR polynomial
3. $\theta(L) = 1 + \theta_1 L + ... + \theta_q L^q$ is the MA polynomial

Regarding VAR, let

$$Y_t = \begin{bmatrix} \log \mathrm{EUI}_t \\ \log \mathrm{GPR}_t \\ \log \mathrm{CPU}_t \\ \log \mathrm{OIL}_t \end{bmatrix}$$

A VAR(p) is:

$$Y_t = c + A_1 Y_{(t-1)} + A_2 Y_{(t-2)} + ... + A_p Y_{(t-p)} + u_t$$

where:

1. c is a vector of intercepts,
2. $A_i$ are coefficient matrices,
3. $u_t$ is a white-noise vector.

The model directly forecasts log(EUI) from its own lags and from the other log-variables.

## 3.2 Bayesian Models

In order to compare models built with different approaches, they must be based on similar theory and implementation.

Regarding the benchmark model, the choice has fallen on a bayesian AR(3), which is a model that bases its prediction only on the previous lags, ignoring any other factor (beside possible dummy variables).

$$y_t = \sum_{j=1}^{p} \phi_j y_{t-j} + \sum_{k=1}^{K} \psi_k d_{kt} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2), \quad t = 1, ..., T$$

where

1. $y_{t-j}$ are the lag values
2. $d_{kt}$ are the (possible) dummy variables
3. $\varepsilon_t$ is the error term

The Bayesian VAR(p), also called BVAR(p), is a model that includes the previous lags of *log_eui* and considers also the endogenous variables presented in the preprocessing section. In fact, they are endogenous, and therefore this is a VAR model, because they are strictly related with each other and an increase in one of them as a direct and clear effect on the target variable.

The formal structure of a BVAR(p) model is the following:

$$Y_t = \Phi' X_t + e_t, \quad e_t \sim N(0, \Sigma), \quad t = 1, \ldots, T$$

where $X_t = (y_{t-1}, \ldots, y_{t-p}, d_t)$ are the lagged values and the dummy variables at time $t$, while $\Phi'$ is the coefficient matrix.

### 3.2.1 Priors

AR(3) and BVAR(2) implement two completely different priors approaches.

Bayesian AR(3) has an informed prior based on the frequentist AR(3), obtained via OLS on the training set [1]. This gives:

1. Coefficient estimates for $\phi_1, \phi_2, \phi_3$ are used as the prior means
2. Standard errors of those coefficients are used as the prior standard deviations
3. Residual standard deviation are used as the scale for $\sigma$

4. The training sample mean of *log_eui* is used as the prior mean for c

All the parameters follow a Normal distribution, while the likelihood follows a t-Student for a higher robustness to outliers.

For the BVAR(2), a simpler prior specification of a more diffuse nature was chosen. Specifically, a prior on two sets of unknowns was placed: the vectorized coefficient matrix $\boldsymbol{\beta}$ and the error covariance matrix $\boldsymbol{\Sigma}$. The diffuse prior assumes:

- $\beta \sim N(\mu_\beta, V_\beta)$: a multivariate Normal on all coefficients;
- $\Sigma \sim IW(\nu_0, \Psi_0)$: an Inverse-Wishart on the error covariance.

Setting $\boldsymbol{\mu}_\beta = \mathbf{0}$ expresses no prior directional belief about any coefficient. Setting $\mathbf{V}_\beta = \mathbf{I}$ gives every coefficient a prior variance of 1, independently of all others. This way, the likelihood dominates and the posterior is driven almost entirely by the data.

With $n = 4$ variables, $\nu_0 = 6 = n + 2$ is the minimum value that keeps the Inverse-Wishart proper and its mean well-defined. The scale matrix $\boldsymbol{\Psi}_0 = \mathbf{I}$ implies a prior mean for $\boldsymbol{\Sigma}$ of

$$E[\boldsymbol{\Sigma}] = \frac{\boldsymbol{\Psi}_0}{\nu_0 - n - 1} = \mathbf{I},$$

expressing no prior knowledge about the error covariance structure.

### 3.3 Parameters Choice

The parameter choices are grounded in a previous project and guided by the trade-off between parsimony and predictive performance. ARIMA(2,0,1) serves as the benchmark: it is the simplest specification capable of producing reliable forecasts, relying exclusively on the lagged values of *log_eui*.
The choice of p = 3 for the Bayesian benchmark may appear inconsistent with the ARIMA(2,0,1). However, this reflects a deliberate modelling decision: a Bayesian AR(2) proved insufficient to capture the full autocorrelation structure of the data. Moreover, the objective of this project is not to find strictly analogous specifications, but to identify which inferential approach delivers the best forecasting performance.
For the VAR, information criteria initially suggested a lag order of 7, but this proved to induce clear overfitting, with out-of-sample performance deteriorating

substantially. VAR(2) was therefore selected as it achieves the best predictive accuracy while generalizing well beyond the training sample, a common finding in multivariate time series, where parsimony often outperforms complexity. The same lag order is then naturally carried over to the BVAR(2), ensuring comparability across specifications.

# 4 Diagnostics

The residual diagnostics (Figure 2) reveal that the ARIMA(2,0,1) model does not produce white-noise residuals: significant autocorrelation persists at medium horizons, indicating that the model fails to fully capture the dynamic structure of *log_eui*. However, this does not disqualify the model as a benchmark and its forecasting performance should be interpreted with this structural shortcoming in mind.

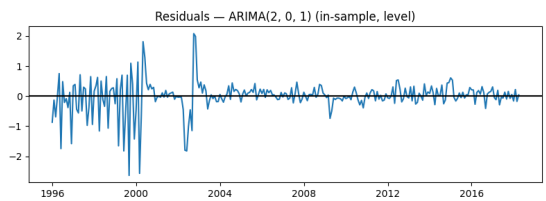


Figure 2: ARIMA(2,0,1) residual diagnostics.

Regarding the Bayesian AR, the convergence diagnostics provide strong evidence that the NUTS sampler operated correctly. The maximum R-hat of 1.0035 confirms that all four chains converged to the same stationary distribution. The Geweke diagnostic passes for all parameters, with the largest absolute z-score of 1.11, confirming that the burn-in period was more than sufficient. Trace plots (Figure 3) show well-mixed chains throughout with no trends or drifts, and running posterior means stabilize within the first 100–200 draws. Even so, the model remains underspecified and insufficient to capture the joint dynamics of the system, which is an expected limitation given that it uses only the lags of *log_eui* as predictors.

Regarding VAR, the residual ACF and PACF (Figure 4) show no persistent autocorrelation across the system's equations, confirming that two lags are sufficient to capture the joint dynamics of the variables. This validates the order selection against the information criteria, which had suggested a VAR(7) at the cost of clear

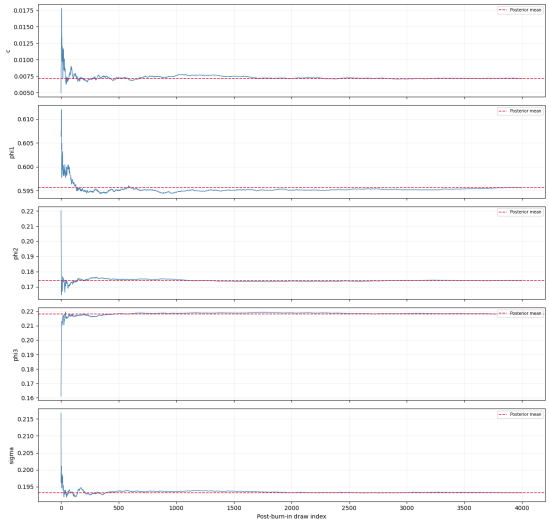


Figure 3: Bayesian AR(3) convergence diagnostics.

overfitting: VAR(2) achieves diagnostic adequacy while preserving the parsimony necessary for reliable out-of-sample performance.

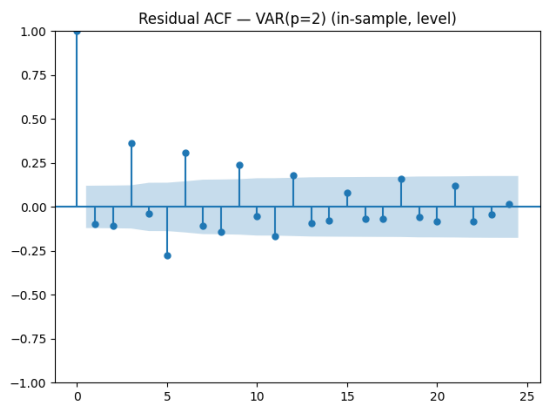


Figure 4: VAR(2) residual diagnostics.

The convergence diagnostics for the Bayesian VAR provide evidence that the Gibbs sampler operated correctly across the full 36-parameter system. The maximum R-hat is 1.0003, confirming that all four independently-seeded chains converged to the same stationary distribution. The Geweke diagnostic passes for all 36 parameters, with a maximum absolute z-score of 1.935. Some of the 36 trace plots are shown below in Figure 5. All the chains stabilize before 1000 draws. The convergence diagnostics for the BVAR(2) are therefore fully satisfactory.

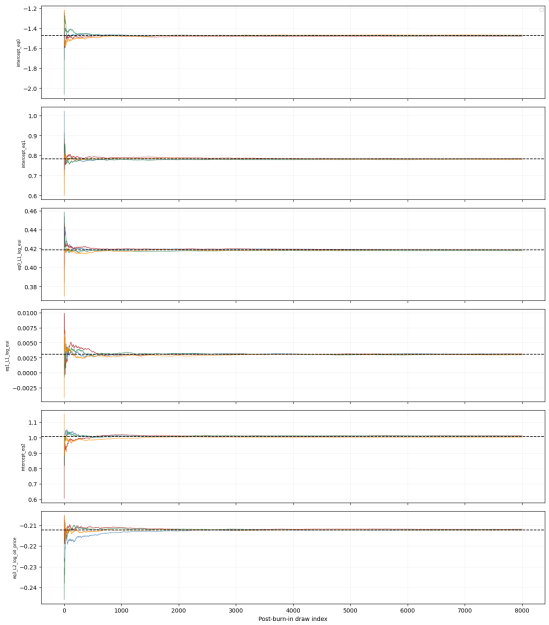


Figure 5: BVAR(2) convergence and residual diagnostics.

# 5 Results Evaluation

The frequentist ARIMA(2,0,1) and the Bayesian AR(3) score almost identical on every point accuracy metric: a RMSE of 0.3530 against 0.3534, and MAE of 0.2364 against 0.2411, differences that could be considered not significant. Taken together, the convergence diagnostics confirm that both estimators are working correctly within their respective frameworks; the bottleneck is the model specification itself. Meaningful forecast improvements are therefore unlikely to come from further refinement of the univariate autoregressive structure.

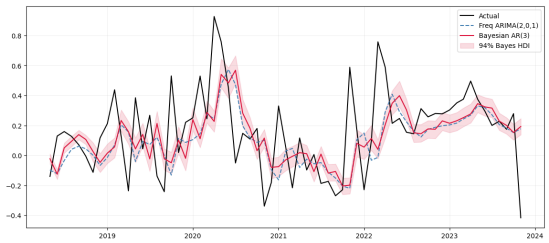


Figure 6: Forecast comparison: ARIMA(2,0,1)-Bayesian AR(3).

The multivariate specifications deliver an improvement over their univariate counterparts. Across all three metrics, BVAR(2) scores better than VAR(2), as shown in Table 1.

| **Model** | **RMSE** | **MAE** | **CRPS** |
|---|---|---|---|
| VAR(2) | 0.3407 | 0.2385 | 0.2139 |
| BVAR(2) | 0.3376 | 0.2307 | 0.2130 |

Table 1: Out-of-sample forecast accuracy.

The margins are modest (approximately 1-2%), but they are consistent across metrics.

All the three metrics capture different effects: RMSE penalises large errors disproportionately, MAE treats all errors symmetrically and CRPS evaluates the entire predictive distribution rather than a single point, rewarding a model that not only centres its forecast correctly but also calibrates its uncertainty appropriately.

Running the Diebold-Mariano Test [2], it is possible to notice that the only significant metric over which the BVAR outperforms VAR is MAE, suggesting that BVAR's advantage is concentrated in reducing moderate-sized errors, rather than large ones.

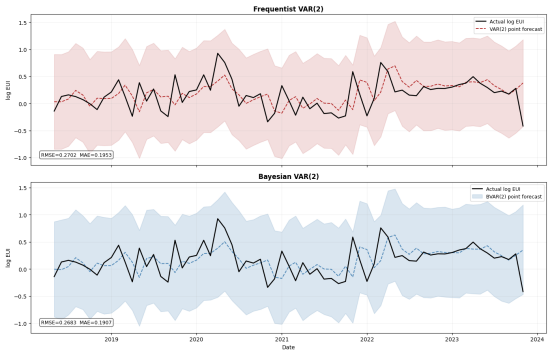


Figure 7: Forecast comparison: VAR(2)-BVAR(2).

In conclusion, the comparison between the univariate benchmark models reveals that the frequentist and Bayesian approaches reach equivalent predictive performance. Neither ARIMA(2,0,1) nor Bayesian AR(3) outperforms the other.
On the other hand, BVAR(2) outperforms VAR(2) consistently across all three metrics. However, the Diebold-Mariano test reveals that this advantage is statistically significant only under MAE, while the differences under RMSE and CRPS

remain indistinguishable from noise. This suggests that the Bayesian framework's gain is concentrated in reducing moderate-sized errors.

# 6 Conclusion

This project set out to evaluate whether a Bayesian inferential framework delivers better forecasting performance relative to the frequentist approach, across both univariate and multivariate specifications.
The broader lesson is that the Bayesian approach does not dominate by construction. Its advantage materializes when the model is more complex. In the sparse univariate case, frequentist and Bayesian estimators converge to the same answer.
Future work could explore several natural extensions. Introducing an informed priors in place of the diffuse specification may amplify the BVAR's advantage.
Finally, extending the framework to a time-varying parameter VAR would allow the cross-variable relationships between geopolitical risk, commodity uncertainty and oil prices to evolve over time.

# References


[1] Robert B Litterman. Forecasting with bayesian vector autoregressions—five years of experience. *Journal of Business & Economic Statistics*, 4(1):25–38, 1986.

[2] Francis X Diebold and Roberto S Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

[3] Gary Koop and Dimitris Korobilis. Bayesian multivariate time series methods for empirical macroeconomics. *Foundations and Trends in Econometrics*, 3(4):267–358, 2010.